Quest 7: The Explouding

Lord of the Wings · impulses spreading (and exploding) through a graph of clouds

> This guide uses the official puzzle text. The story is summarised and the rules are restated in plain words. It explains what the quest asks and the logic behind my solutions, not the code itself. Every number in the examples and traces was checked by running my solutions on the sample data (plus small instrumented copies to print intermediate values).

THE QUESTION

Fire an impulse into one cloud; it spreads along one-way connections, losing 1 strength per hop and adding its strength to every cloud it lands on. Report the total energy at the end.

INPUT

One cloud per line, `id=ID, energy=ENERGY, connections=ID_1,ID_2,...`

OUTPUT

The total energy of all (remaining) clouds.

WHAT CHANGES BETWEEN PARTS

Part I uses strength 5 from the cloud with the most connections. Part II uses strength 10, so the number of impulses explodes and you have to **count** them instead of listing them. Part III adds real explosions: clouds that reach 1,000,000 disappear and fire new strength-10 impulses, and you pick the best starting cloud.

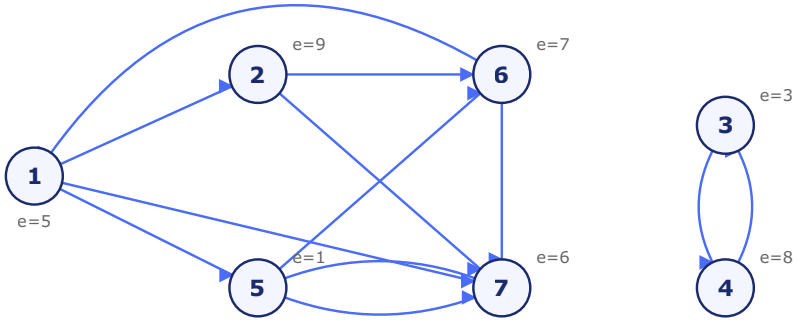
The story

The contamination has reached the whole kingdom's food, so the hero goes to an old Academy friend, Sarumonk, for help with dark magic. Sarumonk is already corrupted, still angry about a prank on his graduation robes, cages the team and demands a lightning duel. To win it, the hero has to work out how charge sent into one cloud spreads across the sky, first on practice notes, then for real, when overloaded clouds start bursting.

Understanding the question

The clouds form a directed graph

Each cloud is a node, and each `connections` entry is a one-way arrow. Arrows can go back to earlier clouds, form cycles (`3 → 4 → 3` in the sample), or even point at the cloud itself.



The sample clouds (Part I, the first Part II example, Part III). Arrows are one-way connections; *e* is the starting energy.

How an impulse spreads

- The first cloud receives strength S (5 in Part I, 10 later) and adds S to its energy.
- Any cloud hit by strength $s > 1$ sends a **separate copy** of strength $s - 1$ along **every** outgoing arrow. A strength-1 impulse adds 1 and stops.
- Copies never merge or cancel. Two impulses arriving at the same cloud each add their energy and each send their own copies. So this is **counting paths**: the energy added to a cloud is the sum, over every path of length k from the start, of $S - k$.
- All impulses move in lockstep: at stage k every live impulse has strength $S - k$. That's why the puzzle lists the energy "by strength".

Extra examples of my own (all run through my solution)

Clouds (all start at energy 0)	Part I (strength 5)	Part II (strength 10)	Why it's interesting
`1 → 1` (a self-loop)	15	55	The impulse keeps landing on cloud 1: 5+4+3+2+1 = 15 and 10+9+…+1 = 55.
`1 → 2`, `2 → 1`	15	55	Same total, split as 5+3+1 = 9 on cloud 1 and 4+2 = 6 on cloud 2.
`1 → 2,3`, `2 → 4`, `3 → 4`, cloud 4 has `connections=` (empty)	19	44	A diamond: cloud 4 is reached along two paths and gets 3 + 3 = 6 in Part I. The empty list is a reading edge case (see below).

> **Empty connection list (lucky behaviour):** `connections=` with nothing after it would be read as one fake connection to cloud 0. That's harmless, because there is no cloud 0 and impulses to it are ignored (the diamond example above relies on this). Your real input has no empty lists.

PART I One impulse of strength 5

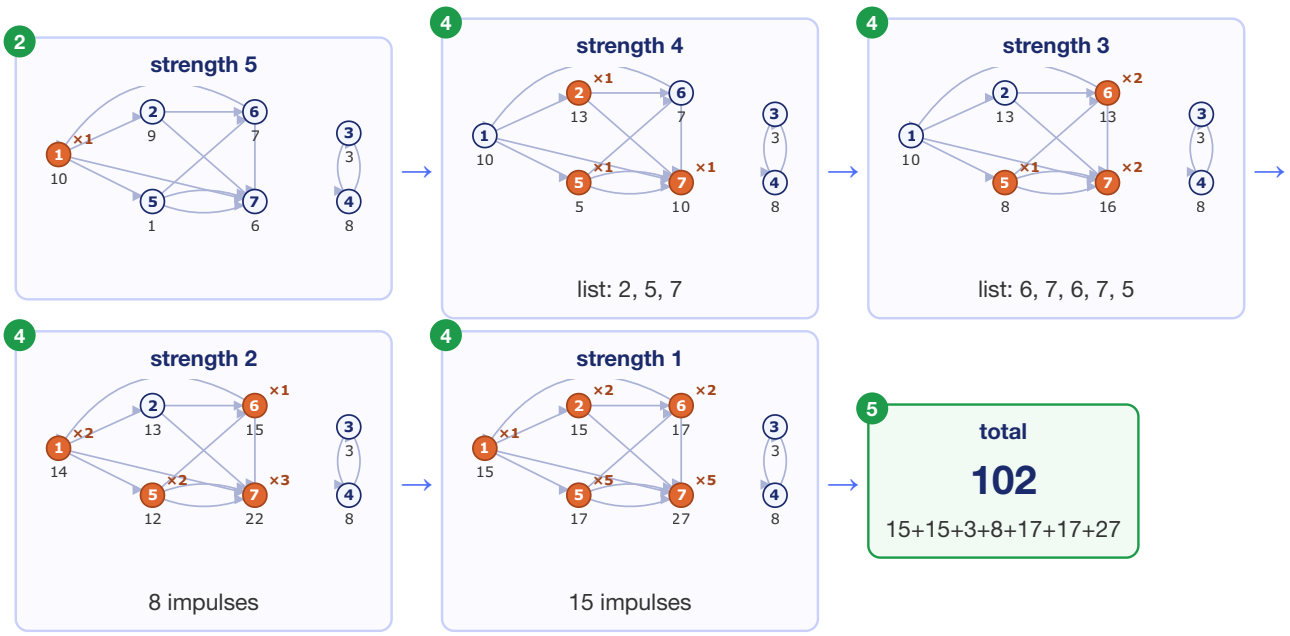
Send strength 5 into the cloud with the most outgoing connections, let it spread completely, and add up the energy of all clouds (including the ones never reached).

Step-by-step solution

1. Read every cloud's id, energy and connection list.
2. Pick the cloud with the most connections and add 5 to it.
3. Make a list of impulses: one entry per connection of the start cloud.
4. For strengths 4, 3, 2, 1: add the strength to the cloud of every impulse in the list, and build the next list from all their connections.
5. Print the sum of all clouds' energy.

How my solution works

Start at the cloud with the most connections (cloud 1) and add 5. Keep a **list of impulses**, one entry per impulse; each stage adds the strength to every impulse's cloud and builds the next list from their connections:



Orange = clouds hit this stage (×n impulses); the number under each cloud is its energy afterwards. Dots match the step numbers above.

> **The list grows fast:** its length is the number of paths (3 → 5 → 8 → 15 here). Fine at strength 5; at strength 10 it wouldn't be, which is what Part II is about.

PART II Strength 10

Same process, but the first impulse has strength 10, so there are ten stages (10, 9, …, 1).

The key idea: group identical impulses

Impulses with the same strength at the same cloud behave identically, so keep a **count per cloud** instead of a list:

Part I: a list (strength 1)

2 5 7 5 6 7 2 5 7 5 6 7 1 7 5

15 entries, one per impulse

=

Part II: a count per cloud

1×1 2×2 5×5 6×2 7×5

5 entries, same information

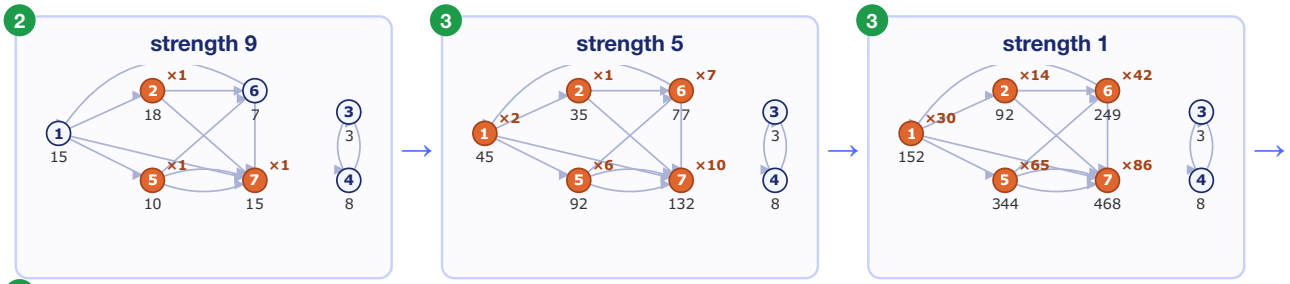
On the denser second example, 134,638 impulses reach cloud 2 in the last stage alone; counted, that's one number.

Step-by-step solution

1. Read the clouds and pick the start as in Part I; add 10 to it.
2. Instead of a list, keep a count of impulses per cloud: start with 1 per connection of the start cloud.
3. For strengths 9 down to 1: each cloud with *n* impulses gains *n* × strength, and adds *n* to the next count of every cloud it connects to.
4. Print the sum of all clouds' energy.

How my solution works

A cloud with *n* impulses gains *n* × strength and sends *n* impulses along each connection. On the Part I clouds:



Orange = clouds hit this stage (×n impulses); the number under each cloud is its energy afterwards. Dots match the step numbers above.

The denser second example gives **1214633**, also matching the puzzle.

PART III Explosions

Now a cloud that reaches 1,000,000 energy **explodes**, in this order:

1. All impulses of the current stage land and pass on their weaker copies as usual (even from clouds about to explode).
2. Every cloud now at or above the limit explodes: it disappears, its energy no longer counts, and it receives nothing ever again.
3. Each exploded cloud sends a fresh **strength-10** impulse along each connection. These travel in the next stage together with the ordinary, weaker impulses.

Try every cloud as the start and report the highest total energy of the clouds that survive. (In the sample, the limit is 100.)

Worked example (sample, limit 100)

Start cloud	1	2	3	4	5	6	7
Total left	108	**187**	94	94	101	11	159

Cloud 1 has the most connections, but starting at cloud 2 is best: **187**.

Extra examples of my own (limit 100, run through a copy of my solution)

Clouds	Start 1	Start 2	What happens
`1 → 2` (e=0), `2 → 1` (e=80)	38	31	Starting at 1, cloud 2 reaches 101 in the strength-5 stage and explodes. Its new strength-10 impulse and the ordinary strength-4 one both land on cloud 1 together in the next stage: 24 + 4 + 10 = 38.
`1 → 2`, `2 → 1` (both e=0)	55	55	10 + 9 + … + 1 = 55 is shared between the two clouds, and neither gets near the limit, so nothing explodes.

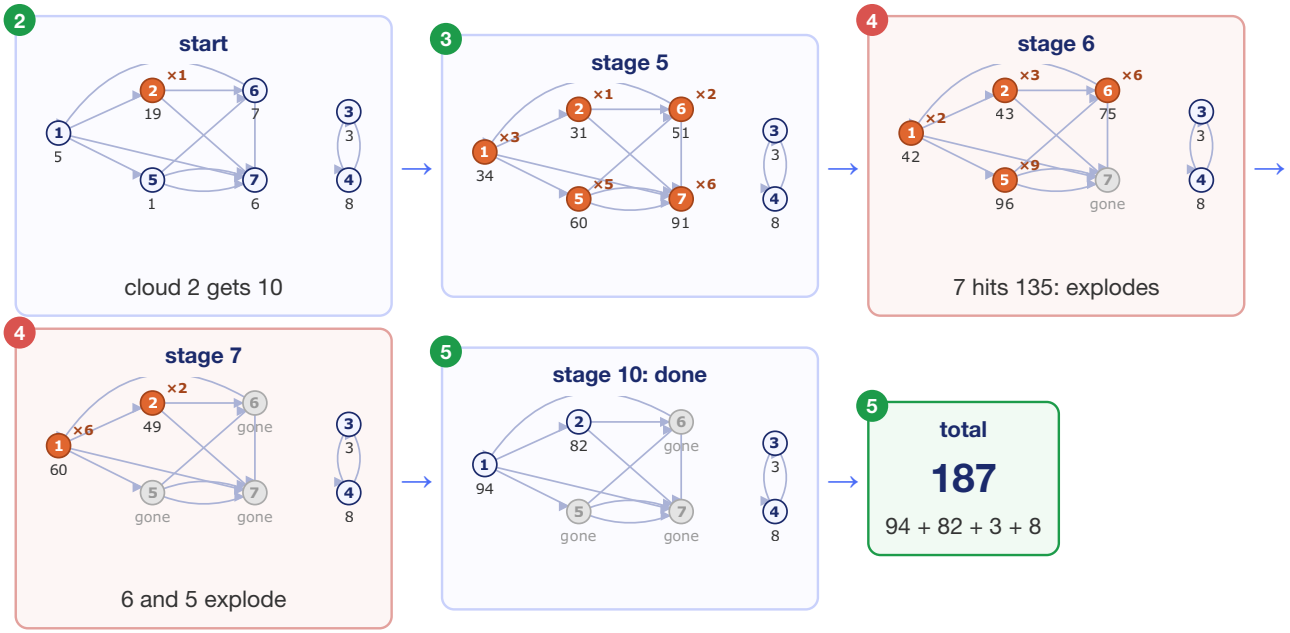
Step-by-step solution

1. Read the clouds. For every cloud as the start, run steps 2–5 on a fresh copy.
2. Add 10 to the start; count strength-9 impulses along its connections, keyed by cloud *and* strength.
3. Each stage: every live cloud with impulses gains strength × count, and (if strength > 1) passes count copies of strength − 1 along its connections.
4. Then every cloud hit this stage that is at or above the limit sends strength-10 impulses along its connections and is removed.

↻ *Repeat until no impulses are left, sum the surviving clouds, and keep the best total over all starts.*

How my solution works

Try **every** cloud as the start, each on a fresh copy of the sky, and keep the best total. After an explosion one stage can hold several strengths, so impulses are counted per **(cloud, strength)**. Each stage: first every group lands and passes on its weaker copies; **then** every hit cloud at the limit explodes and sends strength-10 impulses. Stages run until no impulse is left. Starting at cloud 2:



Orange = clouds hit this stage (×n impulses); the number under each cloud is its energy afterwards. Dots match the step numbers above.

- Stage 6: cloud 7 still passes on its 11 impulses (as strength 3) *before* exploding, then adds one strength-10 impulse to cloud 5.
- Stage 7: cloud 5 gets strength-3 *and* strength-10 impulses at once, which is why counts are kept per (cloud, strength).
- After that, impulses aimed at exploded clouds are dropped; the duel ends after stage 10.

The starting cloud's own +10 is never checked for an explosion. Only clouds hit *during* a stage are checked, so a start cloud pushed over the limit by its first +10 survives. Example (limit 100): `1 → 2` (e=95), `2 → 3`, `3 → 2` (e=1): my solution says 152; by the rules cloud 1 should explode at once, giving 102. Your real input has no cloud within 10 of the limit (checked), so your answer is not affected.

Why a duel ends, and cost: each cloud explodes at most once and impulses lose 1 strength per stage, so at most ten stages follow the last explosion. The whole search runs in about 0.2 s on the real input.

Summary

Part	Start	Strength	Explosions?	How my solution tracks impulses	Sample
I	most connections	5	no	a list with one entry per impulse	102
II	most connections	10	no	a count of impulses per cloud	1316 / 1214633
III	every cloud, keep best	10	yes (limit 1,000,000; 100 in the sample)	a count per (cloud, strength), run until none are left	187

Key ideas: the energy a cloud gets is a sum over **paths**, and paths multiply quickly; impulses that are in the same state (same cloud, same strength) behave identically, so **count them instead of listing them** (the same trick as "lanternfish"-style puzzles); and when the rules say "first everything moves, then everything explodes", process the stage in two passes.